

939.

ON A CASE OF THE INVOLUTION $AF + BG + CH = 0$, WHERE
 A, B, C, F, G, H ARE TERNARY QUADRICS.

[From the *Messenger of Mathematics*, vol. XXII. (1893), pp. 182–186.]

We have here the six conics

$$A = 0, \quad B = 0, \quad C = 0, \quad F = 0, \quad G = 0, \quad H = 0;$$

the curves $AF = 0$ and $BG = 0$ are quartics intersecting in 16 points, and if 8 of these lie in a conic $H = 0$, then the remaining 8 will be in a conic $C = 0$. I take the first set of eight points to be 1, 2, 3, 4, 5, 6, 7, 8; the quartics $AF = 0$ and $BG = 0$ each pass through these eight points; and I assume for the moment

$$A = 1234, \quad F = 5678; \quad B = 1256, \quad G = 3478,$$

viz. that $A = 0$ is a conic through the points 1, 2, 3, 4, and similarly for F, G, B .

Here $H = 0$ is a conic through the points 1, 2, 3, 4, 5, 6, 7, 8, or attending only to the last four points it is a conic through 5, 6, 7, 8; we have therefore a linear relation between F, G, H , and supposing the implicit constant factors to be properly determined, this may be taken to be $F + G + H = 0$; the identity $AF + BG + CH = 0$ thus becomes $F(A - C) + G(B - C) = 0$. We have thus F a numerical multiple of $B - C$, and by a proper determination of the implicit factor we may make this relation to be $F = B - C$; the last equation then gives $G = C - A$, and from the equation $F + G + H = 0$, we have $H = A - B$; the six functions thus are

$$\begin{array}{ll} A, & B - C, \quad \text{or if we please, } A - D, \quad B - C, \\ B, & C - A, \quad \quad \quad B - D, \quad C - A, \\ C, & A - B, \quad \quad \quad C - D, \quad A - B, \end{array}$$

where D is an arbitrary quadric function. The solution

$$(A - D)(B - C) + (B - D)(C - A) + (C - D)(A - B) = 0$$

of the involution is an obvious and trivial one.

But the case which I proceed to consider is

$$A = 1234, \quad F = 5678; \quad B = 1256, \quad G = 3478;$$

here $AF = 0$, and $BG = 0$, meet as before in the points 1, 2, 3, 4, 5, 6, 7, 8, and in eight other points, say that

$$\begin{array}{llll} A = 0, & B = 0 & \text{meet in 1, 2} & \text{and in two other points } \alpha, \beta, \\ A = 0, & G = 0 & ,, & 3, 4 \quad ,, \quad \gamma, \delta, \\ F = 0, & B = 0 & ,, & 5, 6 \quad ,, \quad \epsilon, \zeta, \\ F = 0, & G = 0 & ,, & 7, 8 \quad ,, \quad \eta, \theta; \end{array}$$

then the 8 points $\alpha, \beta, \gamma, \delta, \epsilon, \zeta, \eta, \theta$ will lie in a conic $C = 0$.

I take $y^2 - zx = 0$ for the conic $H = 0$; for any point in this conic we have $x : y : z = 1 : \theta : \theta^2$, and we may take $\theta_1, \theta_2, \theta_3, \theta_4, \theta_5, \theta_6, \theta_7, \theta_8$ for the parameters of the points 1, 2, 3, 4, 5, 6, 7, 8 respectively.

Write $(a, b, c, f, g, h \mid x, y, z)^2 = 0$ for the conic $A, = 1234 = 0$; therefore we have

$$a + b\theta_i + c\theta_i^2 + f\theta_i^3 + g\theta_i^4 + h\theta_i^5 = 0, \quad \theta = \theta_1, \theta_2, \theta_3, \theta_4;$$

or, if

$$\begin{aligned} p_{1234} &= \theta_1 + \theta_2 + \theta_3 + \theta_4, \\ q_{1234} &= \theta_1\theta_2 + \theta_1\theta_3 + \theta_1\theta_4 + \theta_2\theta_3 + \theta_2\theta_4 + \theta_3\theta_4, \\ r_{1234} &= \theta_1\theta_2\theta_3 + \theta_1\theta_2\theta_4 + \theta_1\theta_3\theta_4 + \theta_2\theta_3\theta_4, \\ s_{1234} &= \theta_1\theta_2\theta_3\theta_4, \end{aligned}$$

then

$$c = 1, \quad f = -p_{1234}, \quad b + g = q_{1234}, \quad h = -r_{1234}, \quad a = s_{1234};$$

or, writing $g = -\lambda$, we have

$$s_{1234}x^2 + q_{1234}y^2 + z^2 - p_{1234}yz - r_{1234}xy + \lambda(y^2 - zx) = 0$$

for the equation of the conic in question. We may without loss of generality put $\lambda = 0$; and then if, in general,

$$\Omega = sx^2 + qy^2 + z^2 - pyz - rxy,$$

we have $A = \Omega_{1234} = 0$ for the conic $A = 0$. And thus the equations of the four conics are

$$A = \Omega_{1234} = 0, \quad F = \Omega_{5678} = 0; \quad B = \Omega_{1256} = 0, \quad G = \Omega_{3478} = 0,$$

or, as for shortness I write them,

$$A = \Omega = 0, \quad F = \Omega' = 0; \quad B = \Omega'' = 0, \quad C = \Omega''',$$

viz. in Ω the suffixes are 1, 2, 3, 4, in Ω' they are 5, 6, 7, 8, in Ω'' they are 1, 2, 5, 6, and in Ω''' they are 3, 4, 7, 8.

I find that the implicit constant factors of AF and BG are 1, -1 , and consequently that the form of the identity is

$$\Omega\Omega' - \Omega''\Omega''' + (y^2 - zx)C = 0,$$

where C is a quadric function to be determined; or, what is the same thing, we have

$$\begin{aligned} & (sx^2 + qy^2 + z^2 - pyz - rxy)(s'x^2 + q'y^2 + z^2 - p'yz - r'xy) \\ & - (s''x^2 + q''y^2 + z^2 - p''yz - r''xy)(s'''x^2 + q'''y^2 + z^2 - p'''yz - r'''xy) \\ & + (y^2 - zx)C = 0. \end{aligned}$$

Writing for shortness

$$\begin{aligned} \theta_1 + \theta_2 &= \alpha, & \theta_1\theta_2 &= \beta, \\ \theta_3 + \theta_4 &= \alpha', & \theta_3\theta_4 &= \beta', \\ \theta_5 + \theta_6 &= \alpha'', & \theta_5\theta_6 &= \beta'', \\ \theta_7 + \theta_8 &= \alpha''', & \theta_7\theta_8 &= \beta''', \end{aligned}$$

we have

$$\begin{aligned} p &= \alpha + \alpha', & p' &= \alpha'' + \alpha''', & p'' &= \alpha + \alpha'', & p''' &= \alpha' + \alpha''', \\ q &= \alpha\alpha' + \beta + \beta', & q' &= \alpha''\alpha''' + \beta'' + \beta''', & q'' &= \alpha\alpha'' + \beta + \beta'', & q''' &= \alpha'\alpha''' + \beta' + \beta''', \\ r &= \alpha\beta' + \alpha'\beta, & r' &= \alpha''\beta''' + \alpha'''\beta'', & r'' &= \alpha\beta'' + \alpha''\beta, & r''' &= \alpha'\beta''' + \alpha'''\beta', \\ s &= \beta\beta', & s' &= \beta''\beta''', & s'' &= \beta\beta'', & s''' &= \beta'\beta'''. \end{aligned}$$

In the last-mentioned equation, the first and second lines together are a quartic function of (x, y, z) , say the value is

$$\begin{aligned} & = Ax^4 + By^4 + Cz^4 \\ & + Fy^2z^2 + Gz^2x^2 + Hx^2y^2 \\ & + Iy^3z + Jz^3x + Kx^3y \\ & + Lx^2yz + Mxy^2z + Nxyz^2 \\ & + Py^3x + Qz^3y + Rx^3z, \end{aligned}$$

where after all reductions

$$\begin{aligned}
 A &= ss' - s''s''' = 0, \\
 B &= qq' - q''q''' = -(\alpha\alpha''' - \alpha'\alpha'')(\beta' - \beta'') \\
 &\quad + (\beta'\beta'' - \alpha''\beta')(\alpha - \alpha''') - (\beta' - \beta'')(\beta - \beta'''), \\
 C &= 1 - 1 = 0, \\
 F &= -pq' - p'q + p''q''' + p'''q'', \\
 G &= 0 - 0 = -(\alpha - \alpha''')(\beta' - \beta'') + (\alpha' - \alpha'')(\beta - \beta'''), \\
 H &= -rs' - r's + r''s''' + r'''s'', \\
 I &= -p - p' + p'' + p''', \\
 J &= 0 - 0, \\
 K &= -qr' - q'r + q''r''' + q'''r'' = -(\alpha\beta''' - \alpha''' \beta)(\beta'' - \beta') + (\alpha'\beta'' - \alpha''\beta')(\beta''' - \beta) \\
 &\quad = (\alpha\beta''' - \alpha''' \beta)(\beta - \beta'') + (\alpha'\beta'' - \alpha''\beta')(\beta - \beta'''), \\
 L &= -ps' - p's + p''s''' + p'''s'' = (\alpha\beta''' - \alpha''' \beta)(\alpha'' - \alpha') + (\alpha'\beta'' - \alpha''\beta')(\alpha''' - \alpha), \\
 M &= pr + p'r' - p''r''' - p'''r'' = (\alpha - \alpha''')(\beta'' - \beta') + (\alpha' - \alpha'')(\beta''' - \beta), \\
 N &= -r - r' + r'' + r''' = (\beta' - \beta'')(\beta - \beta'''), \\
 P &= -p'p''' - q'' - q''' = \dots, \\
 Q &= s \dots - s - s = -(\beta - \beta''')(\beta' - \beta''), \\
 R &= rr' + qs' + q's - r''r''' - q''s''' - q'''s'' = 0;
 \end{aligned}$$

values which satisfy

$$F + N = 0, \quad K + L = 0, \quad B + M + Q = 0.$$

The quartic function is thus seen to be

$$= (y^2 - zx)(By^2 + Fyz - Qzx + Kxy) = 0,$$

viz. we have $By^2 + Fyz - Qzx + Kxy = 0$ for the equation of the conic $C = 0$.

Moreover, substituting for p, q, r, s , &c., their values, we have finally for the required involution

$$\begin{aligned}
 &[\beta\beta'x^2 + (\alpha\alpha' + \beta + \beta')y^2 + z^2 - (\alpha + \alpha')yz - (\alpha\beta' + \alpha'\beta)xy] \\
 &\times [\beta''\beta'''x^2 + (\alpha''\alpha''' + \beta'' + \beta''')y^2 + z^2 - (\alpha'' + \alpha''')yz - (\alpha''\beta''' + \alpha'''\beta'')xy] \\
 &- [\beta\beta''x^2 + (\alpha\alpha'' + \beta + \beta'')y^2 + z^2 - (\alpha + \alpha'')yz - (\alpha\beta'' + \alpha''\beta)xy] \\
 &\times [\beta'\beta'''x^2 + (\alpha'\alpha''' + \beta' + \beta''')y^2 + z^2 - (\alpha' + \alpha''')yz - (\alpha'\beta''' + \alpha'''\beta')xy] \\
 &- (y^2 - zx) \left\{ y^2 [(\alpha\beta''' - \alpha''' \beta)(\alpha' - \alpha'') + (\alpha'\beta'' - \alpha''\beta')(\alpha - \alpha''') - (\beta - \beta''')(\beta' - \beta'')] \right. \\
 &\quad + yz [(\alpha - \alpha''')(\beta' - \beta'') + (\alpha' - \alpha'')(\beta - \beta''')] \\
 &\quad - zx [(\beta - \beta''')(\beta' - \beta'')] \\
 &\quad \left. - xy [(\alpha\beta''' - \alpha''' \beta)(\beta' - \beta'') + (\alpha'\beta'' - \alpha''\beta')(\beta - \beta''')] \right\} = 0.
 \end{aligned}$$

It will be recollected that this is the solution for the case $A = 1234, F = 5678; B = 1256, G = 3478$: being that to which the present paper has reference.

940.

ON THE DEVELOPMENT OF $(1 + n^2x)^{\frac{m}{n}}$.

[From the *Messenger of Mathematics*, vol. XXII. (1893), pp. 186–190.]

It is a known theorem that, if $\frac{m}{n}$ be any fraction in its least terms, the coefficients of the development of $(1 + n^2x)^{\frac{m}{n}}$ are all of them integers, or, what is the same thing, that

$$\frac{m \cdot m - n \dots m - (r-1)n}{1 \cdot 2 \dots r} n^r$$

is an integer. The greater part, but not the whole, of this result comes out very simply from Mr Segar's very elegant theorem, *Messenger*, vol. XXII. (1893), p. 59, "the product of the differences of any r unequal numbers is divisible by $(r-1)!!$ " or, as it may be stated, if $\alpha, \beta, \gamma, \dots$ are any r unequal numbers, then $\xi^4(\alpha, \beta, \gamma, \dots)$ is divisible by $\xi^4(0, 1, 2, \dots, r-1)$.

In fact, writing $r+1$ for r and considering the numbers

$$m+n, n, 2n, 3n, \dots, (r-1)n;$$

then neglecting signs

$$\begin{aligned} \xi^4(\alpha, \beta, \gamma, \dots) &= m \cdot m - n \dots m - (r-1)n, \\ &\quad \times 1n \cdot 2n \dots (r-1)n, \\ &\quad \times 1n \cdot 2n \dots (r-2)n, \\ &\quad \vdots \\ &\quad \times 1n \cdot 2n, \\ &\quad \times 1n, \end{aligned}$$

which is

$$= m \cdot m - n \dots m - (r-1)n \times n^{\frac{1}{2}r(r-1)} \times \xi^4(0, 1, 2, \dots, r-1),$$

and similarly

$$\xi^4(0, 1, 2, \dots, r) = 1 \cdot 2 \cdot 3 \dots r \times \xi^4(0, 1, 2, \dots, r-1);$$

so that, omitting the common factor $\xi^4(0, 1, 2, \dots, r-1)$, we have

$$m \cdot m - n \dots m - (r-1)n \cdot n^{\frac{1}{2}r(r-1)} \text{ divisible by } 1 \cdot 2 \cdot 3 \dots r.$$

It thus appears that the fraction

$$\frac{m \cdot m - n \dots m - (r-1)n}{1 \cdot 2 \dots r},$$

when reduced to its least terms, will contain in the denominator only products of powers of the prime factors of n ; and it remains to show that multiplying this by n^r it will become integral, or what is the same thing that

$$\frac{n^r}{1 \cdot 2 \cdot 3 \dots r}$$

in its least terms will not contain in the denominator any prime factor of n .

Considering in succession the prime numbers 2, 3, 5, ..., first the number 2, we see that in the product $1 \cdot 2 \cdot 3 \dots r$, the number of terms divisible by 2 is $= \left(\frac{r}{2}\right)$, the number of terms divisible by 4 is $= \left(\frac{r}{4}\right)$, that by 8 is $= \left(\frac{r}{8}\right)$, and so on, where $\left(\frac{r}{2}\right)$ denotes the integer part of $\frac{r}{2}$, and so in other cases. Hence the product contains the factor 2, with the exponent

$$\left(\frac{r}{2}\right) + \left(\frac{r}{4}\right) + \left(\frac{r}{8}\right) + \dots, \text{ which exponent is less than}$$

$$\frac{r}{2} + \frac{r}{4} + \frac{r}{8} + \dots \text{ ad inf.}$$

is less than r , say it is less than (r) . Similarly for the number 3, the product contains the factor 3, with the exponent

$$\left(\frac{r}{3}\right) + \left(\frac{r}{9}\right) + \left(\frac{r}{27}\right) + \dots,$$

which exponent is less than

$$\frac{r}{3} + \frac{r}{9} + \frac{r}{27} + \dots \text{ ad inf.}$$

is less than $\frac{1}{2}r$, say it is at most $= (\frac{1}{2}r)$; and so it contains the factor 5 with an exponent which is less than $\frac{1}{4}r$, say it is at most $= (\frac{1}{4}r)$, and generally the prime factor p with an exponent which is less than $\frac{1}{p-1}r$: say it is at most $= \left(\frac{1}{p-1}r\right)$.

This is

$$1 \cdot 2 \cdot 3 \dots r = \frac{1}{K} \cdot 2^{(r)} \cdot 3^{(\frac{1}{2}r)} \cdot 5^{(\frac{1}{4}r)} \dots,$$

where K is a whole number. Hence if $n = 2^\alpha 3^\beta 5^\gamma \dots$, we have

$$\frac{n^r}{1 \cdot 2 \cdot 3 \dots r} = K \cdot 2^{r\alpha-(r)} \cdot 3^{r\beta-(\frac{1}{2}r)} \cdot 5^{r\gamma-(\frac{1}{4}r)} \dots,$$

and here for every prime number 2, 3, 5, ... which is a factor of n , that is, for which the corresponding exponent $\alpha, \beta, \gamma, \dots$ is not $= 0$, the exponents $r\alpha - (r)$, $r\beta - (\frac{1}{2}r)$, $r\gamma - (\frac{1}{4}r)$, ... are all of them positive; and thus the fraction in its least terms does not contain in the denominator any prime factor of n : this is the theorem which was to be proved.

Mr Segar's theorem may without loss of generality be stated as follows: if $\beta, \gamma, \dots$ are any $r-1$ unequal positive integers (which for convenience may be taken in order of increasing magnitude), then $\xi^4(0, \beta, \gamma, \dots)$ is divisible by $\xi^4(0, 1, 2, \dots, r-1)$. A proof, in principle the same as his, is as follows:

We have the determinant

$$\begin{vmatrix} 1, & a^\beta, & a^\gamma, & \dots \\ 1, & b^\beta, & b^\gamma, & \dots \\ 1, & c^\beta, & c^\gamma, & \dots \\ \vdots & & & \end{vmatrix} \text{ divisible by } \begin{vmatrix} 1, & a, & a^2, & \dots \\ 1, & b, & b^2, & \dots \\ 1, & c, & c^2, & \dots \\ \vdots & & & \end{vmatrix},$$

viz. the quotient is a rational and integral function of $a, b, c, \dots$ with coefficients which are positive integers; hence putting $a = b = c, \dots = 1$, the quotient will be a positive integer number. Considering the numerator determinant, and for $a, b, c, \dots$ writing therein $1+a, 1+b, 1+c, \dots$ respectively, where $a, b, c, \dots$ are ultimately to be put each $= 0$, the value is

$$= \begin{vmatrix} 1, & 1 + \beta_1 a + \beta_2 a^2 \dots, & 1 + \gamma_1 a + \gamma_2 a^2 + \dots, & \dots \\ & b & & \\ & \vdots & & \end{vmatrix},$$

where $\beta_1, \beta_2, \dots$ denote the binomial coefficients

$$\frac{\beta}{1}, \frac{\beta \cdot \beta - 1}{1 \cdot 2}, \&c.;$$

attending only to the lowest powers of $a, b, c, \dots$ which enter into the formula, this is

$$= \begin{vmatrix} 1, & \dots \\ 1, & \beta_1, & \beta_2, \\ 1, & \gamma_1, & \gamma_2, \\ \vdots & & \end{vmatrix} \begin{vmatrix} 1, & a, & a^2, & \dots \\ 1, & b, & b^2, & \\ 1, & c, & c^2, & \\ \vdots & & \end{vmatrix},$$

or what is the same thing it is

$$= M \begin{vmatrix} 1, & \dots \\ 1, & \beta, & \beta^2, \\ 1, & \gamma, & \gamma^2, \\ \vdots & & \end{vmatrix} \begin{vmatrix} 1, & a, & a^2, & \dots \\ 1, & b, & b^2, & \\ 1, & c, & c^2, & \\ \vdots & & \end{vmatrix} = M \xi^4(0, \beta, \gamma, \dots) \xi^4(0, 1, 2, \dots),$$

where M is a mere number: it will be recollected that in this form, $a, b, c, \dots$ are not the original $a, b, c, \dots$. Putting herein $\beta, \gamma, \dots = 1, 2, \dots$, the denominator determinant is

$$= M \xi^4(0, 1, 2, \dots) \begin{vmatrix} 1, & a, & a^2, & \dots \\ 1, & b, & b^2, & \\ 1, & c, & c^2, & \\ \vdots & & \end{vmatrix},$$

and hence the quotient, which as already seen is an integer number, is equal to $\xi^4(0, \beta, \gamma, \dots) \div \xi^4(0, 1, 2, \dots)$, the theorem in question.

The original theorem as to the form of $(1 + n^2x)^{\frac{m}{n}}$ is a particular case of Eisenstein's very general theorem that, in the development of any algebraical function of x , it is always possible by substituting for x a proper multiple of x , to make all the coefficients integers. It may be remarked that this would not be so if we had only

$$m \cdot m - n \dots m - (r-1)n \cdot n^{\frac{1}{2}r(r-1)}$$

divisible by $1 \cdot 2 \dots r$; for then, writing Nx for x , the form of the coefficient would have been

$$\frac{K N^r}{n^r \cdot n^{\frac{1}{2}r(r-1)}} = \frac{K N^r}{n^{\frac{1}{2}r(r+1)}},$$

and there would be no value (however great) of N by which the denominator factor $n^{\frac{1}{2}r(r+1)}$ could be got rid of.

941.

NOTE ON THE PARTIAL DIFFERENTIAL EQUATION

$$Rr + Ss + Tt + U(s^2 - rt) - V = 0.$$

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. XXVI. (1893), pp. 1–5.]

It is well known that this equation, R, S, T, U, V being any functions whatever of (x, y, z, p, q) , in the case where u admits of an integral of the form $u = f(v)$ (u, v functions of x, y, z, p, q , and f an arbitrary functional symbol) can be integrated as follows; viz. taking m_1, m_2 as the roots of the quadratic equation

$$m^2 - Sm + RT - UV = 0,$$

(that is, writing $m_1 + m_2 = S$ and $m_1 m_2 = RT - UV$), then, m_1 denoting either root at pleasure, and m_2 the other root of the quadratic equation, if the system of ordinary differential equations

$$\begin{aligned} m_1 dx - R dy + U dq &= 0, \\ -T dx + m_2 dy + U dp &= 0, \\ -V dx + m_2 dq + R dp &= 0, \\ -V dy + T dq + m_1 dp &= 0, \\ -p dx - q dy + dz &= 0, \end{aligned}$$

(equivalent to three independent equations) admits of two integrals $u = \text{const.}$ and $v = \text{const.}$, the solution of the given partial differential equation is $u = f(v)$.

In fact, to prove this, we have

$$\begin{aligned} du &= \lambda (m_1 dx - R dy + U dq) \\ &+ \mu (-T dx + m_2 dy + U dp) \\ &+ \nu (-V dx + m_2 dq + R dp) \\ &+ \rho (-V dy + T dq + m_1 dp) \\ &+ \sigma (-p dx - q dy + dz), \end{aligned}$$

that is,

$$\begin{aligned} \frac{du}{dx} &= \lambda m_1 - \mu T - \nu V - \sigma p, \\ \frac{du}{dy} &= -\lambda R + \mu m_2 - \rho V - \sigma q, \\ \frac{du}{dz} &= \sigma, \\ \frac{du}{dp} &= \mu U + \nu R + \rho m_1, \\ \frac{du}{dq} &= \lambda U + \nu m_2 + \rho T, \end{aligned}$$

and thence

$$\frac{du}{dx} + \frac{du}{dz} p + \frac{du}{dp} r + \frac{du}{dq} s = \lambda(m_1 + Ur) + \mu(-T + Us) + \nu(-V + Rr) + \rho(m_1 s + Tt + \dots),$$

$$\frac{du}{dy} + \frac{du}{dz} q + \frac{du}{dp} s + \frac{du}{dq} t = \lambda(-R + Us) + \mu(m_2 + Ut) + \nu(Rs + m_2 t) + \rho(-V + m_1 s + Tt),$$

which equations may be represented by

$$= A\lambda + B\mu + C\nu + D\rho,$$

$$= A'\lambda + B'\mu + C'\nu + D'\rho,$$

and for λ, μ, ν, ρ writing $\lambda', \mu', \nu', \rho'$, we have similarly

$$\frac{d(v)}{dx} = A\lambda' + B\mu' + C\nu' + D\rho',$$

$$\frac{d(v)}{dy} = A'\lambda' + B'\mu' + C'\nu' + D'\rho';$$

whence

$$\frac{d(u)}{dx} \frac{d(v)}{dy} - \frac{d(u)}{dy} \frac{d(v)}{dx} = \begin{vmatrix} A\lambda + B\mu + C\nu + D\rho, & A\lambda' + B\mu' + C\nu' + D\rho' \\ A'\lambda + B'\mu + C'\nu + D'\rho, & A'\lambda' + B'\mu' + C'\nu' + D'\rho' \end{vmatrix}.$$

The determinant is

$$\begin{aligned} &= (AD' - A'D)(\lambda\rho' - \lambda'\rho) + (BD' - B'D)(\mu\rho' - \mu'\rho) \\ &\quad + (CD' - C'D)(\nu\rho' - \nu'\rho) + (BC' - B'C)(\mu\nu' - \mu'\nu) \\ &\quad + (CA' - C'A)(\nu\lambda' - \nu'\lambda) + (AB' - A'B)(\lambda\mu' - \lambda'\mu). \end{aligned}$$

The determinants $AD' - A'D$, &c., each of them contain the factor

$$\Theta_1 = Rr + Ss + Tt + U(s^2 - rt) - V;$$

viz. we have

$$AD' - A'D = m_1\Theta, \quad BC' - B'C = -m_2\Theta,$$

values which give

$$BD' - B'D = -T\Theta, \quad CA' - C'A = -R\Theta,$$

$$CD' - C'D = -V\Theta, \quad AB' - A'B = U\Theta,$$

$$(AD' - A'D)(BC' - B'C) + (BD' - B'D)(CA' - C'A) + (CD' - C'D)(AB' - A'B) = \Theta^2(-m_1m_2 + TR - VU) = 0,$$

as it should be.

Hence, when the partial differential equation $\Theta = 0$ is satisfied, we have

$$\frac{d(u)}{dx} \frac{d(v)}{dy} - \frac{d(u)}{dy} \frac{d(v)}{dx} = 0;$$

and we thence have $u = f(v)$ as the integral of the partial differential equation.

It should be possible to express analytically the conditions in order that the systems of differential equations may have one or each of them two integrals.

It is interesting to remark that, if each of the two systems of ordinary differential equations has only a single integral, these two integrals do not lead to the solution of the partial differential equation. Consider, for instance, the case

$$R = 0, \quad S = x + y, \quad T = 0, \quad U = 0, \quad V = p + q;$$

the partial differential equation is here

$$(x + y)s - (p + q) = 0,$$

which has an integral

$$z = (x + y)\{\phi'(x) + \psi'(y)\} - 2\{\phi(x) + \psi(y)\},$$

where ϕ, ψ are arbitrary functions: the equation in m is $m^2 - m(x + y) = 0$, the roots of which are $m = 0$, and $m = x + y$.

For $m_1 = 0$, $m_2 = x + y$, the system of differential equations becomes

$$\begin{aligned} dy &= 0, \\ -(p + q) dx + (x + y) dq &= 0, \\ -p dx + dz &= 0, \end{aligned}$$

which has only the integral $y = \text{const.}$; and similarly for $m_1 = x + y$, $m_2 = 0$, the system becomes

$$\begin{aligned} dx &= 0, \\ -(p + q) dy + (x + y) dp &= 0, \\ -q dy + dz &= 0, \end{aligned}$$

which has only the integral $x = \text{const.}$ And these two integrals $y = \text{const.}$ and $x = \text{const.}$ do not in anywise lead to the integral of the partial differential equation.

I take the opportunity of remarking that the complete system of conditions in order that the differential

$$A dx + B dy + C dz + D dw$$

may be $= M dU$ is as follows: viz. writing

$$A, B, C, D = 1, 2, 3, 4;$$

$$\frac{dB}{dz} - \frac{dC}{dy}, \frac{dC}{dx} - \frac{dA}{dz}, \frac{dA}{dy} - \frac{dB}{dx}, \frac{dA}{dw} - \frac{dD}{dx}, \frac{dB}{dw} - \frac{dD}{dy}, \frac{dC}{dw} - \frac{dD}{dz} = 23, 31, 12, 14, 24, 34,$$

where of course $12 = -21$, &c., and

$$123 = 1 \cdot 23 + 2 \cdot 31 + 3 \cdot 12, \text{ \&c.}; \quad 1234 = 1 \cdot 234 - 2 \cdot 341 + 3 \cdot 412 - 4 \cdot 123,$$

is $= 0$ identically;

$$1234 = 12 \cdot 34 + 13 \cdot 42 + 14 \cdot 23,$$

then the conditions equivalent to three independent conditions are

$$234 = 0, \quad 341 = 0, \quad 412 = 0, \quad 123 = 0, \quad 1234 = 0.$$

In fact, the first four equations are

$$\begin{aligned} 2 \cdot 34 - 3 \cdot 24 + 4 \cdot 23 &= 0, \\ -1 \cdot 34 + 3 \cdot 14 + 4 \cdot 31 &= 0, \\ 1 \cdot 24 - 2 \cdot 14 + 4 \cdot 12 &= 0, \\ -1 \cdot 23 - 2 \cdot 31 - 3 \cdot 12 &= 0; \end{aligned}$$

hence, multiplying by 1, 2, 3, 4 respectively and adding, we have the identity $\overline{1234} = 0$, so that these four are equivalent to three independent equations: and multiplying by

$$\begin{aligned} 12 \cdot \mu - 31 \cdot \nu + 14 \cdot \rho, \\ -12 \cdot \lambda + 23 \cdot \nu + 24 \cdot \rho, \\ 31 \cdot \lambda - 23 \cdot \mu + 34 \cdot \rho, \\ -14 \cdot \lambda - 24 \cdot \mu - 34 \cdot \nu \end{aligned}$$

respectively, (where λ, μ, ν, ρ are arbitrary), we have

$$(1 \cdot \lambda + 2 \cdot \mu + 3 \cdot \nu + 4 \cdot \rho)(23 \cdot 14 + 31 \cdot 24 + 12 \cdot 34) = 0,$$

that is,

$$23 \cdot 14 + 31 \cdot 24 + 12 \cdot 34 = 0, \text{ or } 1234 = 0,$$

the fifth condition.

942.

ON SEMINVARIANTS.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. XXVI. (1893), pp. 66–69.]

I wish to prove the following negative: a given sharp seminvariant is not in every case obtainable by mere derivation from a form of the same extent and of the next inferior degree. The meaning of the statement will be explained.

According to the general theory developed in Clebsch's *Theorie der binären algebraischen Formen*, Leipzig, 1872, the covariants of a given binary quantic f are all of them obtainable, the covariants of a given degree from those of the next inferior degree, by derivation (Ueberschiebung) of these with f ; viz. if the covariants of the next inferior degree are P, Q , &c., then the covariants of the degree in question are all of them included among the forms

$$(f, P)^0 (= fP), (f, P)^1, (f, P)^2, \dots, \\ (f, Q)^0, (f, Q)^1, (f, Q)^2, \dots,$$

&c.,

the index of derivation for (f, P) being at most equal to the degree of f or to that of P , whichever of these is the smaller, and so for Q , &c. The forms thus obtained are far too numerous; but rejecting repetitions, we have a complete system of the covariants of the given degree, viz. every covariant whatever of that degree is a linear function (with numerical multipliers) of the several distinct forms thus obtained by derivation.

We can therefore, by linear combination as above, obtain all the sharp covariants of the given degree, but we may very well have a sharp covariant not included among the several distinct forms thus obtained by derivation, but only expressible as a linear combination of two or more such forms; or say we may very well have a sharp covariant not obtainable by mere derivation from the forms of the next inferior degree; and it is important to verify that there are sharp covariants not thus obtainable by mere derivation. I remark that the notion of a sharp covariant does not present itself in Clebsch, and that it will be presently explained.

Passing from a covariant to its leading coefficient which is a seminvariant, the statement may be applied to seminvariants; viz. it is to be verified that there are sharp seminvariants not obtainable by mere derivation from the seminvariants of the next inferior degree. But we have to introduce the notion of "extent" so as to connect the seminvariant with a quantic of some particular order, thus, if the highest letter of the seminvariant is f , we say that the extent is = 5, and thus connect it with the quintic

$$(1, b, c, d, e, f \mid x, y)^5.$$

The notion "sharp" applies to the seminvariants of a given weight. Suppose, for instance, the weight is = 8; we have a series of initial or non-unitary terms i, eg, df, e^2 , &c., and a series of final or power-ending terms e^2, cd^2, b^2d^2, c^4 , &c., and we denote by $eg - c^4$ (where observe that here and in all similar cases the $-$ is not a minus sign, but is simply a stroke), the whole series of terms (including eg and c^4) which are in counter-order (CO) subsequent to eg , and in alphabetical order (AO) precedent to c^4 ; and so in other cases. This being so, arranging the seminvariants with their final terms in AO, we have the seminvariants

$$i - e^2, \\ eg - cd^2, \\ df - b^2d^2, \\ e^2 - c^4,$$

&c.,

viz. we have a seminvariant $i - e^2$ containing all or any (in fact, all) of the terms of this set as above defined; a seminvariant $eg - cd^2$ containing all or any of the terms of this set, a seminvariant $df - b^2d^2$ containing all or any of the terms of this set; and so on. These are sharp forms; a seminvariant ending in e^2 , must of necessity have the leading term i , and thus belong at least to the octic

$$(1, b, c, d, e, f, g, h, i \mid x, y)^8,$$

a seminvariant ending in cd^2 must of necessity have a leading term as high as eg , and thus belong at least to the sextic

$$(1, b, c, d, e, f, g \mid x, y)^6.$$

Any linear combination of these would be a seminvariant $i - cd^2$, belonging to the octic, but it is not a sharp form; the final term cd^2 does not of necessity imply an initial so high as i (in fact, as we have seen, it only implies the lower initial eg); and so in other cases.

For the quintic $(1, b, c, d, e, f \mid x, y)^5$, we have (for the weight 8 and degree 4) the seminvariants $df - b^2d^2$, and $e^2 - c^4$, this last belongs, of course, also to the quartic $(1, b, c, d, e \mid x, y)^4$; it is, in fact, the squared quadrinvariant $(e - 4bd + 3c^2)^2$. I wish to show that $df - b^2d^2$ is not obtainable by mere derivation from the covariants of degree 3 of the quartic.

The quintic and its covariants up to the degree 3 are

$$A = (1, 5b, 10c, 10d, 5e, f \mid x, y)^5,$$

say

$$(1, b, c, d, e, f \mid x, y)^5;$$

$$B = \left(\begin{array}{c|c|c|c} e+1 & f-1 & bf+4 & \\ bd-4 & be-3 & ce-4 & \\ c^2+3 & cd+2 & d^2-3 & \end{array} \mid x, y \right)^2;$$

say

$$(e - c^2, f - cd, bf - d^2 \mid x, y)^2;$$

$$C = (c - b^2, d - bc, e - c^2, f - cd, bf - d^2, cf - de, df - e^2 \mid x, y)^6;$$

$$D = (ce - c^3, cf - c^2d, df - cd^2, bdf - d^3 \mid x, y)^3;$$

$$E = (f - bc^2, bf - c^3, cf - c^2d, df - cd^2, ef - d^3, f^2 - d^2e \mid x, y)^5;$$

$$F = (d - b^3, e - b^2c, f - bc^2, bf - c^3, cf - c^2d, df - cd^2, ef - d^3, f^2 - d^2e, bf^2 - de^2, cf^2 - e^3 \mid x, y)^9.$$

Hence all the covariants of the degree 3 are A^3, AB, AC, D, E, F , where

$$AC = (e - c^2, f - bc^2, bf - c^3, cf - c^2d, bcf - cd^2, bdf - d^3, bef - d^2e, bf^2 - d^2 \mid x, y)^7$$

and the derivatives are

$(A, A^3)^0, (A, A^3)^1, \&c.$	weights of leading coefficients are 0, 1, 2, 3, 4, 5
$(A, AB)^0, (A, AB)^1, \&c.$	4, 5, 6, 7, 8, 9
$(A, AC)^0, (A, AC)^1, \&c.$	2, 3, 4, 5, 6, 7
$(A, D)^0, (A, D)^1, \&c.$	6, 7, 8, 9
$(A, E)^0, (A, E)^1, \&c.$	5, 6, 7, 8, 9, 10
$(A, F)^0, (A, F)^1, \&c.$	3, 4, 5, 6, 7, 8.

The terms giving rise to a seminvariant of weight 8 are

$$(A, AB)^4 = \left\{ \begin{array}{cccccc} e - c^2, & f - bc^2, & bf - c^3, & cf - c^2d, & bcf - cd^2 & \\ e, & d & c, & b & & 1 \end{array} \right\} = df - c^4,$$

$$(A, D)^2 = \left\{ \begin{array}{ccc} ce - c^3, & cf - c^2d, & df - cd^2 \\ d, & c, & b \end{array} \mid 1 \right\} = df - c^4,$$

$$(A, E)^3 = \left\{ \begin{array}{cccc} f - bc^2, & bf - c^3, & cf - c^2d, & df - cd^2 \\ d, & c, & b, & 1 \end{array} \right\} = df - c^4,$$

$$(A, F)^5 = \left\{ \begin{array}{cccccc} d - b^3, & e - b^2c, & f - bc^2, & bf - c^3, & cf - c^2d, & df - cd^2 \\ f, & e, & d, & c, & b, & 1 \end{array} \right\} = df - c^4,$$

where to explain the algorithm, I remark, that if $A = (A_0, A_1, A_2, \dots \mid x, y)^5$ and $D = (D_0, D_1, D_2, D_3 \mid x, y)^3$, represented as above by

$$(A, D)^2 = \left\{ \begin{matrix} D_0, & D_1, & D_2, \\ A_2, & A_3, & A_4 \end{matrix} \right\} = \{ce - c^3 - 2D_3A_3 + D_3A_2 - 2D_2A_3 + D_2A_4\}.$$

The result is in every case given as $df - c^4$; in each case there is only a single term $c \cdot c^3 = c^4$, and the term in c^4 certainly presents itself. In $(A, AB)^4$ there is a single term $d \cdot f = df$, and in $(A, D)^2$ a single term df , and thus the term df certainly presents itself: in $(A, E)^3$ there are two terms $d \cdot f = df$ and df , and it is conceivable that, inserting the proper numerical coefficients, these might destroy each other: if this were so, the form instead of being $df - c^4$ would be $e^2 - c^4$; and similarly in $(A, F)^5$, there are two terms $df = df$ and df , which it is conceivable might destroy each other, and the form would then be $e^2 - c^4$. But in every case we have the term c^4 , and it thus appears that the form $df - b^2d^2$ is not obtainable by mere derivation.

The form in question is in fact obtained by a linear combination of $df - c^4$ and $e^2 - c^4$, viz. writing down the leading coefficients of the covariants B^2 and H , we have

	$3H$	$-2E^2$	$=$
df	3		3
e^2	0	-2	-2
bcf	-9		-9
bde	-15	+16	+1
c^2e	+30	-12	-18
b^2f	-9		-9
	-15		-15
	+6		+6
	0		0
b^2ce	-48		-48
bc^2d	+42	-32	+106
b^2d^2	-15		-15
+18			
c^3d	-18	+48	0
c^4	+18	-18	0

viz. the form in question $df - b^2d^2$ is $= 3df - 2e^2 - \dots - 10b^2d^2$.

943.

ON RECIPROCATS AND DIFFERENTIAL INVARIANTS.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. XXVI. (1893), pp. 169–194, 289–307.]

I use the term Reciprocant to denote a function of an arbitrary variable or variables and its differential coefficients, not connected with any differential equation; and Differential Invariant to denote a function of the coefficients of a differential equation and the differential coefficients of these coefficients. Halphen's differential invariants are thus reciprocants, but the term reciprocant is not made use of by him. I have entitled the present paper "On Reciprocants and Differential Invariants"; in the earlier part, (except that for preserving a chronological order, I briefly refer to Sir J. Cockle's Criticoids, which are differential invariants or rather seminvariants), I attend almost exclusively to Reciprocants, reproducing and explaining, and in some parts developing, the theories of Ampère, Halphen in his first two memoirs, and Sylvester.

I.

The notion of a reciprocant first presents itself in Ampère's "Mémoire sur les avantages qu'on peut retirer dans la théorie des courbes de la considération des paraboles osculatrices, avec des réflexions sur les fonctions différentielles dont la valeur ne change pas lors de la transformation des axes," *Journ. École*

Polyt. t. VII. (1808), pp. 151–191 (sent to the Institute, Dec. 1803).¹ We have (p. 167) for the radius of curvature the expression

$$-\frac{(1+y'^2)^{\frac{3}{2}}}{y''},$$

and for the parameter of the osculating parabola the expression

$$-\frac{5y'^5}{\{y''^{-\frac{2}{3}} + 3(5y''^2/5y'''^3 - 2/y''')\}};$$

and it is explicitly noticed that each of these expressions remains absolutely unaltered when the coordinates x, y are changed into any other (rectangular) coordinates whatever.

The functions here spoken of are thus orthogonal absolute reciprocants; the change which leaves them unaltered is that of x, y into X, Y , where

$$X = x \cos \theta + y \sin \theta + \alpha, \quad Y = -x \sin \theta + y \cos \theta + \beta.$$

For the more general change

$$X = ax + by + \alpha, \quad Y = cx + dy + \beta,$$

we find without difficulty

$$Y' = \frac{c + dy'}{a - by'}, \quad Y'' = \frac{(ad - bc)y''}{(a - by')^3},$$

and thus there is not an identity of form in the expressions

$$\frac{(1+Y'^2)^{\frac{3}{2}}}{Y''} \text{ and } \frac{(1+y'^2)^{\frac{3}{2}}}{y''}.$$

But if we disregard factors, or (what is the same thing) attend to the equations $y'' = 0$ and $Y'' = 0$, we see that for this more general change one of these equations implies the other; each of them is, in fact, the condition for an inflexion of the curve in (x, y) or (X, Y) .

II.

The differential invariants, or rather seminvariants, called “Criticoids,” were considered by Sir James Cockle in his paper “On Criticoids,” *Phil. Mag.* t. XXXIX. (1870), pp. 201–221; these will be more particularly considered further on, but at present it is sufficient to remark that the functions in question are connected with a linear differential equation, viz. they are functions of the coefficients of a linear differential equation, such that effecting thereon a transformation of the dependent variable, say $y = f \cdot Y$, where f is an arbitrary function of the independent variable x , the function is equal to the like function of the coefficients of the new equation. For instance, if by the transformation in question $y = f \cdot Y$ we transform

$$\frac{d^2y}{dx^2} + 2b \frac{dy}{dx} + cy = 0 \text{ into } \frac{d^2Y}{dX^2} + 2B \frac{dY}{dX} + CY = 0,$$

then we have

$$B = b + \frac{f'}{f}, \quad C = c + 2b \frac{f'}{f} + \frac{f''}{f},$$

and thence

$$C - B^2 - B' = c - b^2 - b',$$

¹A reciprocant, the Schwarzian derivative, occurs in Lagrange’s memoir, “Sur la construction des cartes géographiques,” *Nouv. Mém. de Berlin*, 1779, ((Euvres, t. iv. p. 651, but scarcely qua reciprocant, viz. the form in question $\frac{f'''}{f'} - \frac{3}{2} \left(\frac{f''}{f'} \right)^2$, presents itself in the equation

$$\frac{f'''(u+ti)}{f'(u+ti)} - \frac{3}{2} \left(\frac{f''(u+ti)}{f'(u+ti)} \right)^2 = \frac{F'''(u-ti)}{F'(u-ti)} - \frac{3}{2} \left(\frac{F''(u-ti)}{F'(u-ti)} \right)^2,$$

where F is the conjugate function of f , an equation implying that each side thereof is equal to one and the same constant; $u - ti$ is of course not a function of $u + ti$, and thus we have here no property that the function $\frac{f'''}{f'} - \frac{3}{2} \left(\frac{f''}{f'} \right)^2$ remains unaltered when for the arguments x, y thereof we substitute X, Y , determinate functions of these arguments.

(where the accents denote differentiation in regard to x): and thus $c - b^2 - b'$ is an invariant of the form in question; say it is an a -seminvariant of the differential equation.

III.

We have four important Memoirs by Halphen, (1) “Thèse d’Analyse sur les invariants différentiels,” Paris (1878), (2) “Sur les invariants différentiels des courbes gauches,” *Jour. École Polyt. Cah. XLVII.* (1880), pp. 1–102; (3) “Mémoire sur la réduction des équations différentielles linéaires aux formes intégrables,” *Mém. Sav. Étrang.* t. XXVIII. (1883), pp. 1–297, and (4) “Sur les invariants des équations différentielles linéaires du quatrième ordre,” *Acta Math.* t. III. (1883), pp. 325–380.

In the Memoir, Halphen (1), the investigations are in the first instance presented in a geometrical form; the author considers for instance the inflexions of a plane curve, and so obtains the invariant y'' , or using his notation

$$\frac{1}{1 \cdot 2 \dots k} \frac{d^k y}{dx^k} = a_k,$$

say the invariant a_2 of the weight 2; similarly, the consideration of the sextactic points gives him the invariant $a_2^2 a_5 - 3a_2 a_3 a_4 + 2a_3^3$ of the weight 9; and that of the points of nine-pointic contact a more complicated invariant of the weight 27: and with these he forms absolute invariants. But the formal analytical definition is given §3, “Théorie des invariants différentiels jusqu’au huitième ordre exclusivement,” viz. considering the general homographic transformation

$$X = \frac{\alpha x + \beta y + \gamma}{\alpha'' x + \beta'' y + \gamma''}, \quad Y = \frac{\alpha' x + \beta' y + \gamma'}{\alpha'' x + \beta'' y + \gamma''},$$

then if the equation

$$f\left(x, y, \frac{dy}{dx}, \dots, \frac{d^n y}{dx^n}\right) = 0$$

is unaltered in form by the change (x, y) into (X, Y) , or what is the same thing if the function

$$f\left(x, y, \frac{dy}{dx}, \dots, \frac{d^n y}{dx^n}\right)$$

be unaltered save as to a factor, then such function is said to be a differential invariant: and the quotient of two functions having the same factor, which quotient is therefore unaltered, is said to be an absolute invariant. We may for these terms substitute reciprocant, and absolute reciprocant respectively. The theory is scarcely worked out from the definition, and indeed, as will presently appear, it is by no means easy to work out directly from the definition, even the foregoing sextactic invariant $a_2^2 a_5 - 3a_2 a_3 a_4 + 2a_3^3$. I remark that, in what follows, instead of Halphen’s $a_2, a_3, a_4, a_5, \dots$ I write $a, b, c, d, \dots$, viz. these are used to denote respectively

$$\frac{1}{2} \frac{d^2 y}{dx^2}, \frac{1}{6} \frac{d^3 y}{dx^3}, \frac{1}{24} \frac{d^4 y}{dx^4}, \frac{1}{120} \frac{d^5 y}{dx^5}, \dots,$$

the first differential coefficient $\frac{dy}{dx}$ being denoted by t . The foregoing invariant, or say the sextactic reciprocant, is thus $= a^2 d - 3abc + 2b^3$. (See post, XI.)

IV.

For the analytical theory, we have as above

$$X = \frac{\alpha x + \beta y + \gamma}{\alpha'' x + \beta'' y + \gamma''}, \quad Y = \frac{\alpha' x + \beta' y + \gamma'}{\alpha'' x + \beta'' y + \gamma''};$$

taking h, k for the increments of x, y respectively, we have

$$k = th + ah^2 + bh^3 + ch^4 + dh^5 + \dots,$$

and, similarly, taking H, K for the increments of X, Y respectively, we have

$$K = TH + AH^2 + BH^3 + CH^4 + DH^5 + \dots,$$

and we require the relations between the two sets of coefficients $(t, a, b, c, d, \dots)$ and $(T, A, B, C, D, \dots)$: I propose to develop these up to d, D , so as to obtain the theory for the form $a^2 d - 3abc + 2b^3$.

Writing for shortness ξ, η, ζ for

$$\alpha x + \beta y + \gamma, \quad \alpha' x + \beta' y + \gamma', \quad \alpha'' x + \beta'' y + \gamma''$$

respectively, we have

$$X + H = \frac{\alpha(x+h) + \beta(y+k) + \gamma}{\alpha''(x+h) + \beta''(y+k) + \gamma''} = \frac{\xi + \alpha h + \beta k}{\zeta + \alpha'' h + \beta'' k},$$

and thence

$$H = \frac{\xi + \alpha h + \beta k}{\zeta + \alpha'' h + \beta'' k} - \frac{\xi}{\zeta} = \frac{(\alpha\zeta - \alpha''\xi)h + (\beta\zeta - \beta''\xi)k}{\zeta(\zeta + \alpha'' h + \beta'' k)} = \frac{Ph + Qk}{\zeta(\zeta + \alpha'' h + \beta'' k)},$$

if

$$P = \alpha\zeta - \alpha''\xi, \quad Q = \beta\zeta - \beta''\xi,$$

or, for k substituting its value,

$$H = \frac{(P + Qt)h + Q(ah^2 + bh^3 + ch^4 + dh^5)}{\zeta[1 + \lambda''\{h + \lambda''(ah^2 + bh^3 + ch^4 + dh^5)\}]},$$

if

$$L = P + Qt, \quad \lambda = \frac{Q}{P + Qt},$$

$$L'' = \alpha'' + \beta''t, \quad \lambda'' = \frac{\beta''}{\alpha'' + \beta''t}.$$

Similarly, if

$$\frac{Y + K}{X + H} = \frac{\eta + \alpha'h + \beta'k}{\zeta + \alpha''h + \beta''k},$$

then

$$K = \frac{L'h + L'\lambda'(ah^2 + bh^3 + ch^4 + dh^5)}{1 + L''\{h + \lambda''(ah^2 + bh^3 + ch^4 + dh^5)\}};$$

we have thus found H and K each of them in terms of h , and the elimination of h leads to an equation between H and K expressible in the form

$$K = TH + AH^2 + BH^3 + CH^4 + DH^5.$$

Writing for convenience

$$ah^2 + bh^3 + ch^4 + dh^5 = h\Omega,$$

that is,

$$\Omega = ah + bh^2 + ch^3 + dh^4,$$

we have

$$\zeta H = \frac{Lh(1 + \lambda\Omega)}{1 + L''h(1 + \lambda''\Omega)},$$

and thence

$$\zeta K = \frac{L'h(1 + \lambda'\Omega)}{1 + L''h(1 + \lambda''\Omega)},$$

$$K = \zeta H \frac{L'(1 + \lambda'\Omega)}{L(1 + \lambda\Omega)} = TH \frac{1 + \lambda'\Omega}{1 + \lambda\Omega},$$

since, when h is small or say $H = 0$, we have $K = TH$; viz. we thus have $T = \frac{L'}{L}$.

Developing, we have

$$K = TH[1 + (\lambda' - \lambda)\Omega - \lambda(\lambda' - \lambda)\Omega^2 + \lambda^2(\lambda' - \lambda)\Omega^3 - \dots].$$

We have

$$h = \frac{1}{a} \Omega - \frac{b}{a^2} \Omega^2 + \frac{1}{a^3} (2b^2 - ac) \Omega^3 - \frac{1}{a^4} (5b^3 - 5abc + a^2d) \Omega^4 + \dots = p\Omega + q\Omega^2 + r\Omega^3 + s\Omega^4,$$

suppose, and hence

$$\begin{aligned} \zeta H &= \frac{L(p\Omega + q\Omega^2 + r\Omega^3 + s\Omega^4)(1 + \lambda\Omega)}{1 + L''(p\Omega + q\Omega^2 + r\Omega^3 + s\Omega^4)(1 + \lambda''\Omega)} \\ &= \frac{Lp\Omega + L(q + \lambda p)\Omega^2 + L(r + \lambda q)\Omega^3 + L(s + \lambda r)\Omega^4}{1 + L''p\Omega + L''(q + \lambda''p)\Omega^2 + L''(r + \lambda''q)\Omega^3 + L''(s + \lambda''r)\Omega^4}, \end{aligned}$$

which gives H in terms of Ω , and conversely Ω in terms of H . We assume

$$K = TH + AH^2 + BH^3 + CH^4 + DH^5.$$

K is then given as above in terms of H , Ω , that is, in terms of H , and it is assumed that we have

In the foregoing expression for K , substituting for Ω its value

$$XH + YH^2 + ZH^3 + WH^4,$$

and comparing coefficients, we find

$$\begin{aligned} A &= (\lambda' - \lambda) X, \\ B &= (\lambda' - \lambda) Y - \lambda X^2, \\ C &= (\lambda' - \lambda) Z - \lambda \cdot 2XY + \lambda^2 X^3, \\ D &= (\lambda' - \lambda) W - \lambda(2XZ + Y^2) + \lambda^2 \cdot 3X^2Y - \lambda^3 X^4; \end{aligned}$$

and we have then, from the relation between H , Ω , to find the values of X , Y , Z , W . We have

$$\begin{aligned} &p(XH + YH^2 + ZH^3 + WH^4) \\ &+ (q + \lambda p)(X^2H^2 + 2XYH^3 + (2XZ + Y^2)H^4) \\ &+ (r + \lambda q)(X^3H^3 + 3X^2YH^4) \\ &+ (s + \lambda r)X^4H^4 \\ &- pH(XH + YH^2 + ZH^3) \\ &- pH L''(q + \lambda''p)(X^2H^2 + 2XYH^3) \\ &- pH L''(r + \lambda''q)X^3H^3 = 0. \end{aligned}$$

Equating coefficients, we have

$$\begin{aligned} pX - p &= 0, \\ pY + (q + \lambda p)X^2 - pL''pX &= 0, \\ pZ + (q + \lambda p)2XY + (r + \lambda q)X^3 - pL''pY - pL''(q + \lambda''p)X^2 &= 0, \\ pW + (q + \lambda p)(2XZ + Y^2) + (r + \lambda q)3X^2Y + (s + \lambda r)X^4 - pL''pZ - pL''(q + \lambda''p)2XY - pL''(r + \lambda''q)X^3 &= 0, \end{aligned}$$

and substituting for p, q, r, s their values, we find successively after reductions, which for W are somewhat troublesome, the values

$$\begin{aligned} X &= \mu a, \\ Y &= \mu^2 (b + aL'' - a^2\lambda''), \\ Z &= \mu^3 (c + 2bL'' + aL''^2 - (4ac + 2b^2)\lambda'' + 3a^2b\lambda''^2), \\ W &= \mu^4 [d + 3cL'' + 3bL''^2 + aL''^3 + 10a^5\lambda''Z'' - \dots], \end{aligned}$$

these values are to be substituted for X, Y, Z, W in the foregoing expressions of A, B, C, D .

I add the formulæ

$$\lambda' - \lambda = \frac{Q'}{P' + Q't} - \frac{Q}{P + Qt} = \frac{PQ' - P'Q}{(P + Qt)(P' + Q't)},$$

$$PQ' - P'Q = (\alpha\zeta - \alpha''\xi)(\beta'\zeta - \beta''\eta) - (\beta\zeta - \beta''\xi)(\alpha'\zeta - \alpha''\eta),$$

$$= \zeta \begin{vmatrix} \alpha, & \alpha', & \alpha'' \\ \beta, & \beta', & \beta'' \\ \xi, & \eta, & \zeta \end{vmatrix} = \zeta \begin{vmatrix} \alpha, & \alpha', & \alpha'' \\ \beta, & \beta', & \beta'' \\ \gamma, & \gamma', & \gamma'' \end{vmatrix},$$

$$P + Qt = \alpha\zeta - \alpha''\xi + (\beta\zeta - \beta''\xi)t = \alpha(\alpha''x + \beta''y + \gamma'') - \alpha''(\alpha x + \beta y + \gamma) + t\{\beta(\alpha''x + \beta''y + \gamma'') - \beta''(\alpha x + \beta y + \gamma)\},$$

$$Q't = \alpha'\xi - \alpha''\eta + (\beta'\xi - \beta''\eta)t,$$

$$\text{that is, } = \alpha'(\alpha''x + \beta''y + \gamma'') - \alpha''(\alpha'x + \beta'y + \gamma') + t\{\beta'(\alpha''x + \beta''y + \gamma'') - \beta''(\alpha'x + \beta'y + \gamma')\},$$

$$P' + Q't = \begin{vmatrix} \alpha', & \beta', & \gamma' \\ \alpha'', & \beta'', & \gamma'' \\ x, & -1, & y - tx \end{vmatrix}, \quad P + Qt = \begin{vmatrix} \alpha, & \beta, & \gamma \\ \alpha'', & \beta'', & \gamma'' \\ x, & -1, & y - tx \end{vmatrix},$$

whence $\lambda' - \lambda$ is known.

Also

$$\mu = \frac{\zeta}{L} = \frac{\alpha''x + \beta''y + \gamma''}{P + Qt}, \quad \mu' = \frac{\zeta}{L'} = \frac{\alpha''x + \beta''y + \gamma''}{P' + Q't},$$

and

$$T = \frac{\mu}{\mu'} = \frac{L'}{L} = \frac{P' + Q't}{P + Qt}.$$

V.

We have $A = (\lambda' - \lambda)\mu a$, and thus a is a reciprocant, viz. $a = 0$ is the condition of an inflexion.

I remark that $ac - b^2$ is not a reciprocant; we have

$$AC - B^2 = (\lambda - \lambda')^2 (XZ - Y^2),$$

and then

$$XZ - Y^2 = \mu^5 \{(ac - b^2) - a^2\lambda''(b - a^2\lambda'') + a^3(\lambda'' - \lambda)L''\},$$

and thus $AC - B^2$ is not a multiple of $ac - b^2$, even in the particular case $L'' = 0$. Notice further that we have

$$\begin{aligned} (4AC - 5B^2) &= (\lambda' - \lambda)^2 [4XZ - 5Y^2 + 2\lambda X^2Y - \lambda^2 X^4], \\ &= (\lambda' - \lambda)^2 \mu^4 \{4ac - 5b^2 - 2abL'' - a^2L''^2 + 4a^3\lambda''L''\}, \end{aligned}$$

viz. in the particular case $L'' = 0$, we have

$$4AC - 5B^2 = (\lambda' - \lambda)^2 \mu^4 (4ac - 5b^2),$$

and thus in the particular case $L'' = 0$ (or say, if $\alpha'' = 0$, $\beta'' = 0$, that is, if X, Y are mere linear functions of x, y) we have $4ac - 5b^2 = 0$ a reciprocant. But in the general case now in hand, it is not a reciprocant.

We have now to consider the sextactic form $a^2d - 3abc + 2b^3$. We have

$$A^2D - 3ABC + 2B^3 = (\lambda' - \lambda)^3 \{X^2W - 3XYZ + 2Y^3 + \lambda X^2(XZ - Y^2)\},$$

and we proceed to calculate the expression in $\{ \}$. Writing for shortness

$$X = \mu a,$$

$$Y = \mu^2 (b + aL'' + Y_0),$$

$$Z = \mu^3 (c + 2bL'' + aL''^2 + Z_0),$$

$$W = \mu^4 (d + 3cL'' + 3bL''^2 + aL''^3 + W_0),$$

and then, omitting a factor μ^6 , the expression is

$$\begin{aligned} &= a^2d - 3abc + 2b^3 \\ &\quad + a^2W_0 - 3a(b + aL'')Z_0 + \{(-3ac + 6b^2) + 6abL'' + 3a^2L''^2\}Y_0 \\ &\quad + 6(b + aL'')Y_0^2 - 3aY_0Z_0 + 2Y_0^3 \\ &\quad + \lambda a^2\{ac - b^2 + aZ_0 - 2(b + aL'')Y_0 - Y_0^2\}, \end{aligned}$$

where the terms after the first term are in fact $= 0$; and this being so, we have

$$A^2D - 3ABC + 2B^3 = (\lambda' - \lambda)^3 \mu^6 (a^2d - 3abc + 2b^3),$$

and thus $a^2d - 3abc + 2b^3$ is a reciprocant; but observe that the factor being here $(\lambda' - \lambda)^3 \mu^6$, and in the equation $A = (\lambda' - \lambda) \mu a$ the factor being $(\lambda' - \lambda) \mu$, we cannot with the reciprocants a and $a^2d - 3abc + 2b^3$ form an absolute reciprocant.

To verify the evanescence of the above-mentioned terms, observe that, considering first the terms independent of L'' , we have

$$\begin{aligned} a^2W_0 &= -4a^3c\lambda'' - 2a^2b^2\lambda'' + 10a^4b\lambda''^2 - 5a^6\lambda''^3, \\ -3abZ_0 &= +9a^2b^2\lambda'' - 6a^4b\lambda''^2, \\ (-3ac + 6b^2)Y_0 &= +3a^3c\lambda'' - 6a^2b^2\lambda'', \\ -3aY_0Z_0 &= -9a^4b\lambda''^2 + 6a^6\lambda''^3, \\ +6bY_0^2 &= +6a^4b\lambda''^2, \\ +2Y_0^3 &= -2a^6\lambda''^3, \\ +\lambda a^2(ac - b^2) &= +a^3c\lambda - a^2b^2\lambda, \\ +\lambda a^3Z_0 &= -3a^4b\lambda''^2 + 2a^6\lambda''^3, \\ +\lambda a^2 \cdot -2bY_0 &= +2a^4b\lambda''^2, \\ +\lambda a^2 \cdot -Y_0^2 &= -a^6\lambda''^3; \end{aligned}$$

the sum of which is in fact $= 0$. And next for the terms containing L'' , we have

$$\begin{aligned} a^2W_0 &= -12a^3b\lambda L'' + 3a^3b\lambda''L'' - 6a^4\lambda L''^2 + 3a^4\lambda''L''^2 + 10a^5\lambda^2L'' - \dots, \\ -3abZ_0 &= +9a^3b\lambda L'' - 3a^3b\lambda''L'', \\ (6abL'' + 3a^2L''^2)Y_0 &= +9a^3b\lambda L'' + 9a^4\lambda L''^2 - 3a^4\lambda''L''^2 - 3a^4\lambda L''^2, \\ -3aY_0 \cdot L'' &= -6a^3b\lambda L'' + 3a^5\lambda\lambda''L'', \\ +\lambda a^3Z_0 &= +6a^4bL'', \\ -2\lambda a^2(b + aL'')Y_0 &= \dots, \end{aligned}$$

the sum of which is also $= 0$.

Before going further, it is proper to remark that $a, b, c, \dots$ as representing the second, third, &c., differential coefficients of y are considered as being of the orders $2, 3, 4, \dots$, and the order of a reciprocant is taken to be the order of the highest letter contained therein: the degree means the degree in these letters $a, b, c, \dots$, and the weight the weight in these letters, reckoning them as of the weights $2, 3, 4, \dots$. A reciprocant is a homogeneous isobaric function of $a, b, c, \dots$, not involving y or the first differential coefficient.

VI.

To the reciprocants thus obtained, say

$$U = a, \quad V = a^2d - 3abc + 2b^3,$$

Halphen adds another reciprocant Δ (that of nine-pointic contact) which he expresses in the form of a determinant.

Writing down in connexion therewith its developed expression, we have

$$\Delta = \begin{vmatrix} b, & c, & d, & e, & f \\ a, & b, & c, & d, & e \\ -a^2, & 0, & b^2, & 2bd + c^2, & 2be + 2cd \\ 0, & a^2, & 2ab, & 2ac + b^2, & 2ad + 2bc \\ 0, & 0, & a^2, & 3ab, & 3ac + 3b^2 \end{vmatrix},$$

which works out to

$$\begin{aligned} &= a^6(df + 1) \\ &\quad - a^6 \cdot 1 \\ &\quad + a^5(bcf - 3) \\ &\quad + a^5(c^2e - 1) \\ &\quad + a^5(bde + 3) \\ &\quad + a^5(cd^2 - 5) \\ &\quad + a^4(b^3f + 2) \\ &\quad + a^4(b^2ce - 5) \\ &\quad + a^4(bc^2d + 14) \\ &\quad + a^4(c^4 - 4) \\ &\quad + a^3(b^2c^3 - 10) \\ &\quad + a^2(bc^4 + 15) \\ &\quad + a^1(b^4c - 12) \\ &\quad + a^0(b^6 + 3), \end{aligned}$$

where observe that the whole term in the highest letter f is $a^4(a^2d - 3abc + 2b^3)f$, viz. the coefficient of f is the reciprocant $a^4(a^2d - 3abc + 2b^3)$, agreeing with a general theorem given by Halphen.

Δ is of the degree 8 and the weight 24, and it is seen without difficulty that the factor is $= (\lambda' - \lambda)^8 \mu^{16}$. He remarks that $25\Delta^3 - 27V^8$ vanishes for $a = 0$ (viz. it